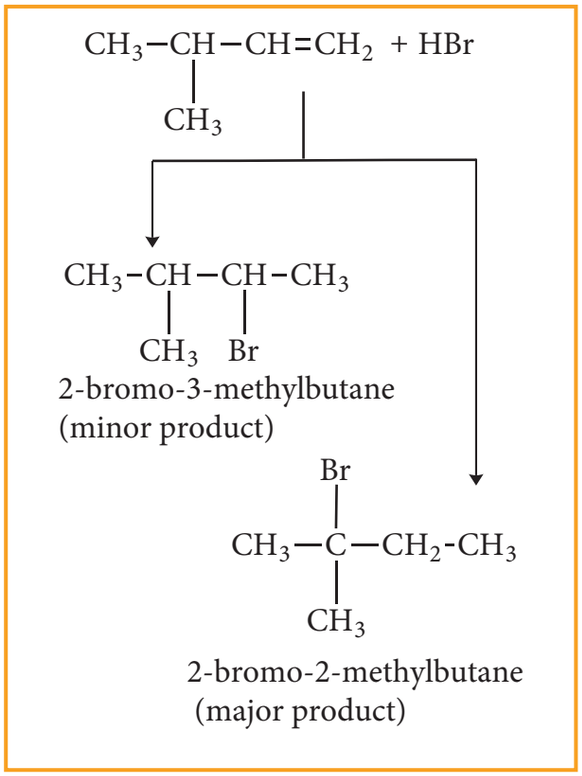
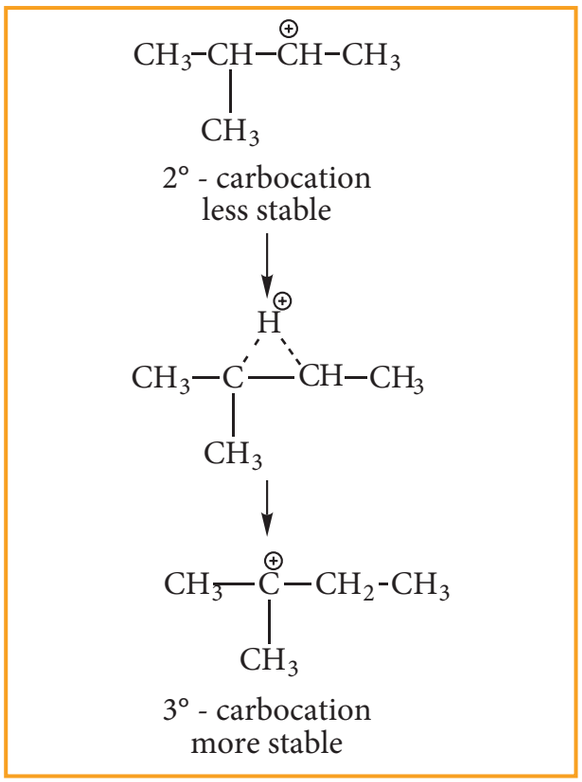
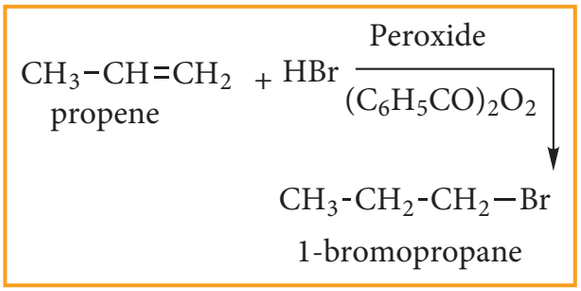


Carbocation rearrangement



Anti-Markovnikoff's Rule (Or) Peroxide Effect (Or) KharaschAddition

The addition of HBr to an alkene in the presence of organic peroxide, gives the anti Markovnikoff's product. This effect is called peroxide effect.

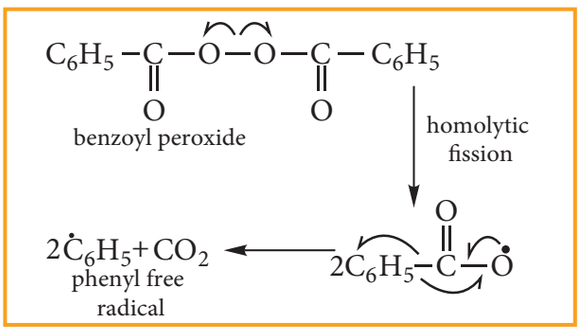


Mechanism:

The reaction proceeds via free radical mechanism.

Step:1

The weak O-O single bond linkages of peroxides undergoes homolytic cleavage to generate free radical.



Step:2

The radicals abstracts a hydrogen from HBr thus generating bromine radical.



Step:3

The Bromine radical adds to the double bond in the way to form more stable alkyl free radical.

Unit 13 - jagan.indd 196 22-01-2021 12:51:34